SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

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Predictive Modeling of PLGA Nanoparticle Properties for Small Molecule Drug Delivery using Machine Learning.

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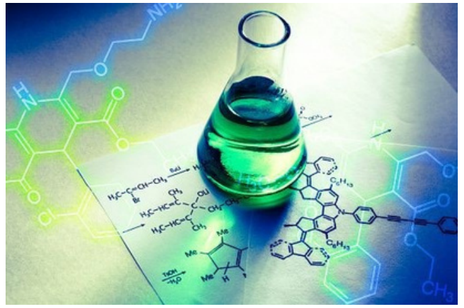


Problem with Drugs

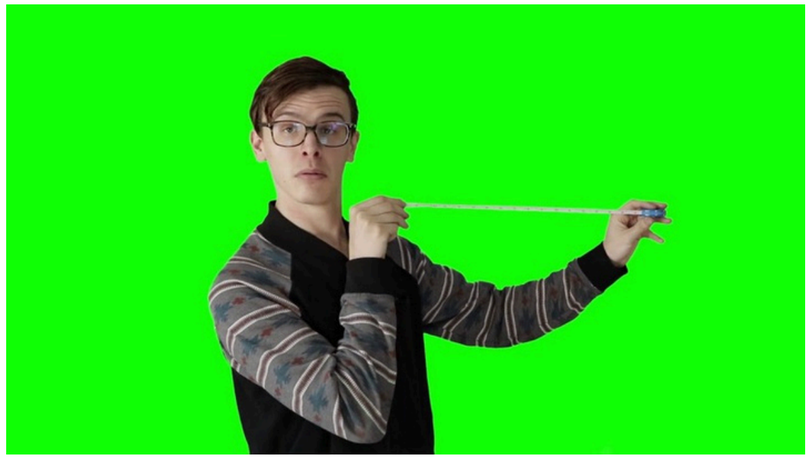
- Low solubility
- Poor targeting
- Side effects
- Degradation in body

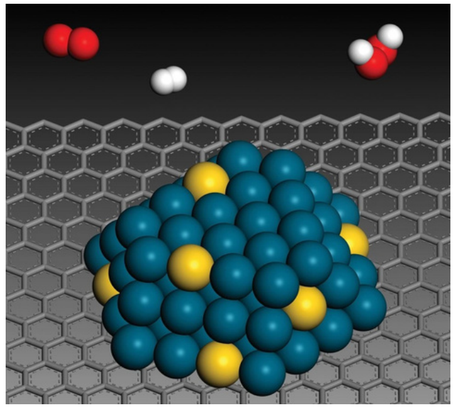
What are Nanoparticles?

- Very small particles (nanometer scale)
- Used as drug carriers

PLGA=Poly(lactide-co-glycolide)

- Biodegradable polymer
- Safe for body
- Used to make nanoparticles





Dataset Overview

- 433 formulations
- 15 input features 3 outputs:
- Particle Size
- EE

P	mol_TPSA	mol_melting_point	mol_Hacceptors	mol_Hdonors	mol_heteroatoms	drug/polymer	surfactant_concentration	surfactant_HLB	aqueo
08	100.59	267.5	6	3	9	1	1.5	18	
08	100.59	267.5	6	3	9	0.5	1.5	18	
08	100.59	267.5	6	3	9	0.2	1.5	18	
08	100.59	267.5	6	3	9	0.1	1.5	18	
08	100.59	267.5	6	3	9	0.05	1.5	18	
15	32.67	132	2	0	4	0.029	0.2	20.5	
15	32.67	132	2	0	4	0.143	0.2	20.5	

Tools & Techniques Used

Libraries Used:

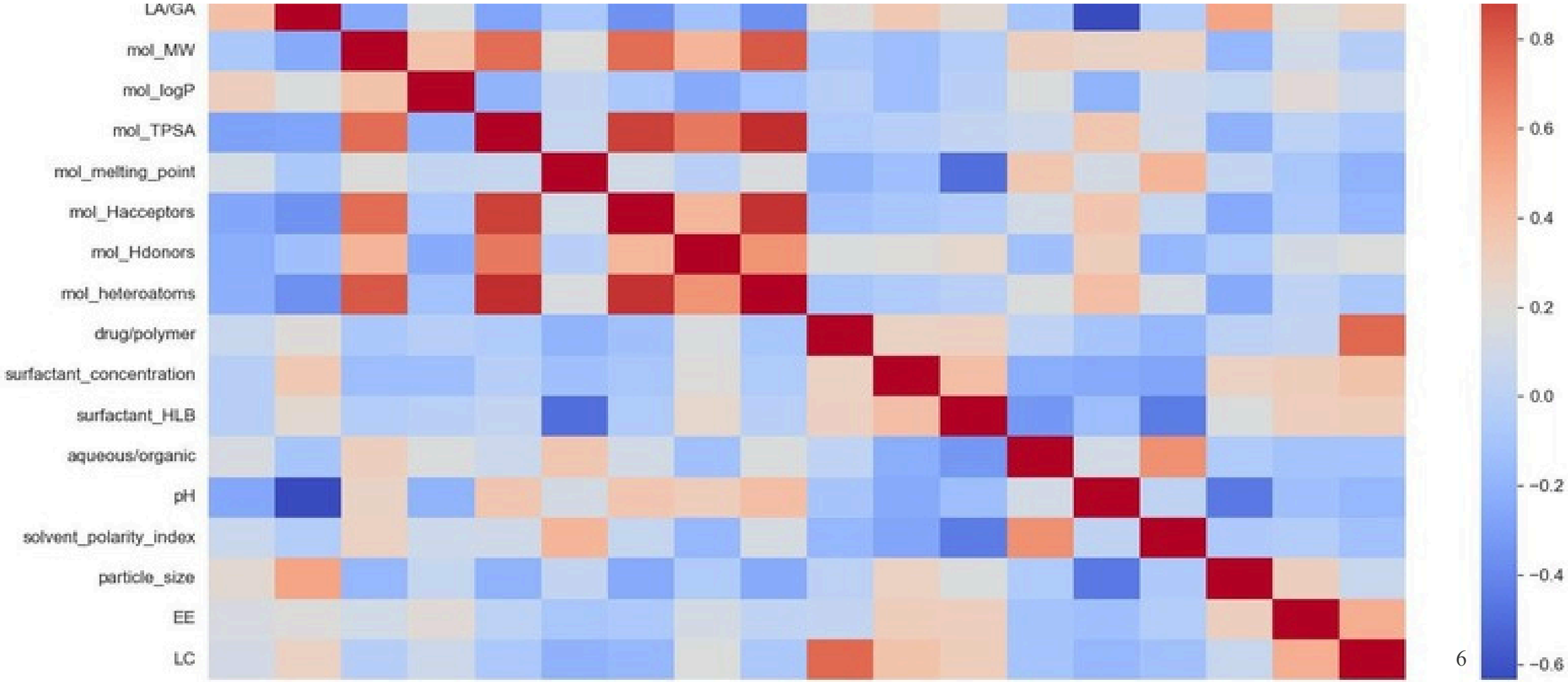
- Pandas → Data handling
- NumPy → Numerical operations
- Matplotlib & Seaborn → Visualization
- Scikit-learn → Machine Learning

Techniques Used:

- Exploratory Data Analysis (EDA)
- Correlation Analysis
- Data Visualization
- Random Forest Regression

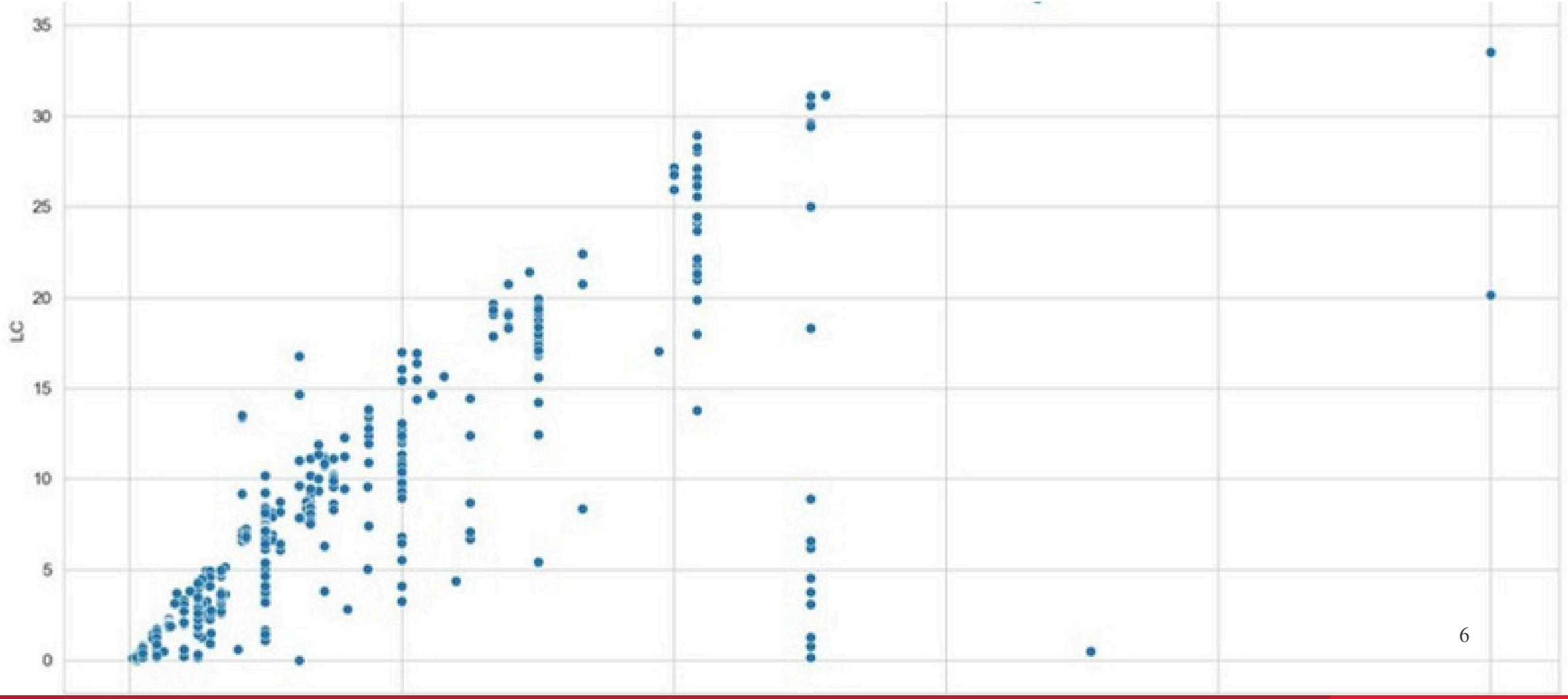
Correlation Heatmap

Relationship between variables



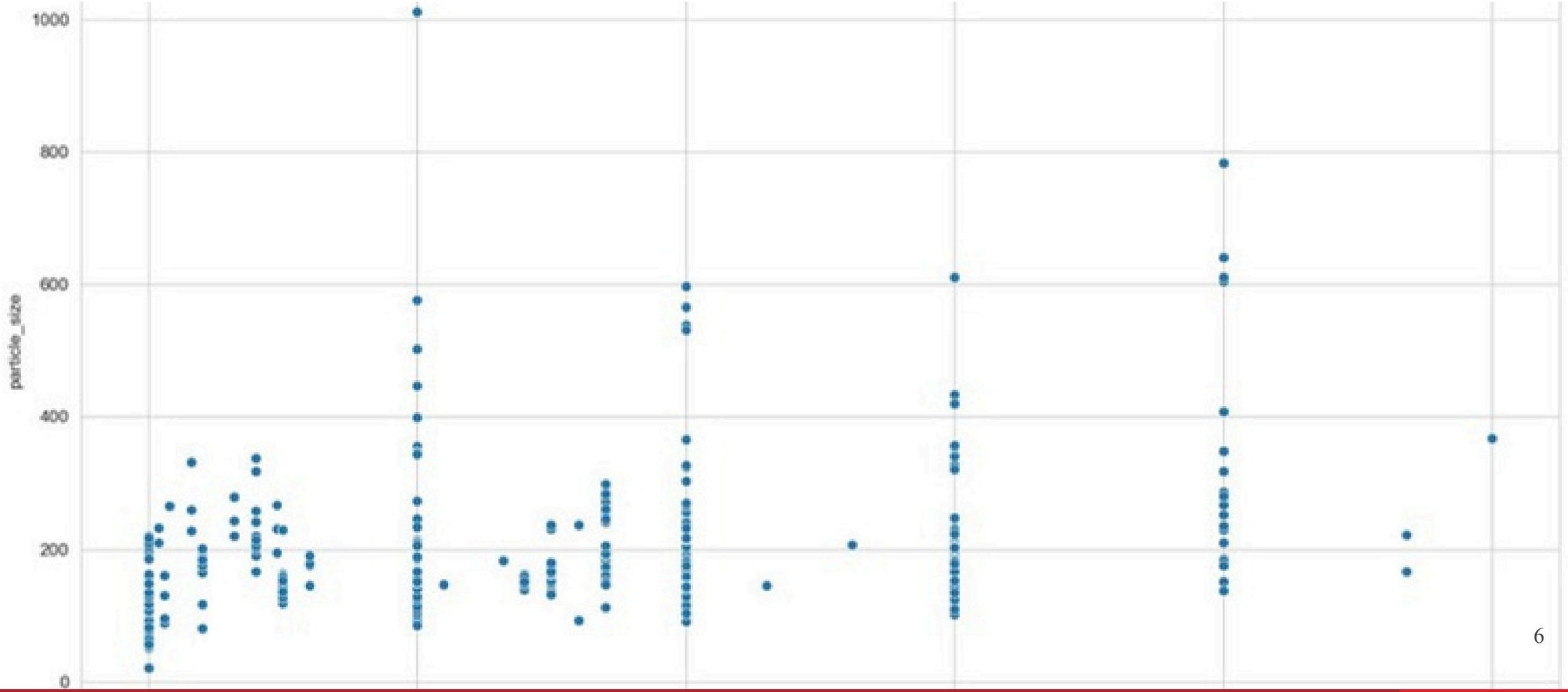
Drug/Polymer vs LC

↑ Drug/Polymer → ↑ LC

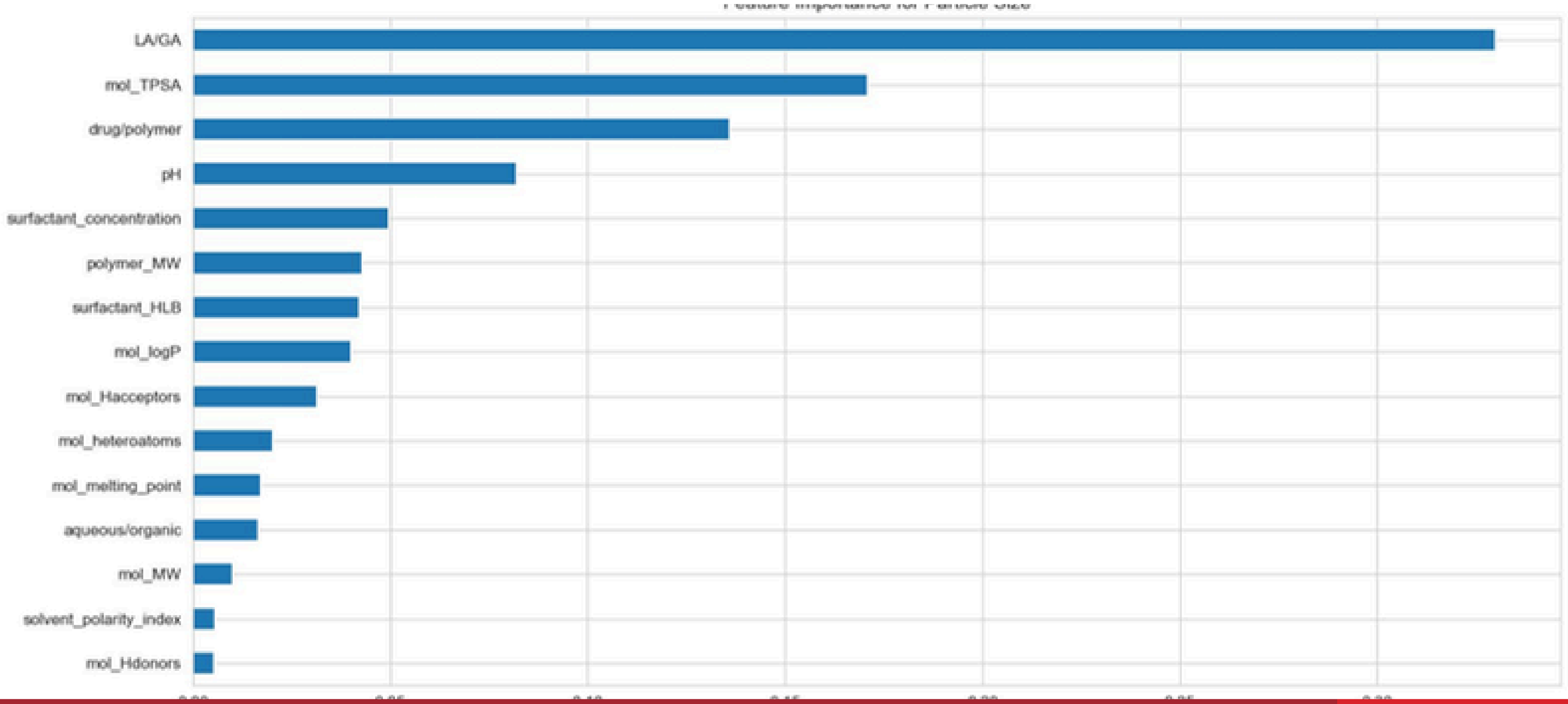


Surfactant vs Particle Size

Surfactant affects size



Key influencing parameters identified



References

- Scientific Data Journal (PLGA dataset paper)
- Mendeley Data Repository
- Scikit-learn Documentation
- Research articles on PLGA nanoparticles

Github repo: https://github.com/SWATVSS107/Predictive-Modeling-of-PLGA-Nanoparticle-Properties/tree/main

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Thank You